

Reconstruction of Human Body Pose and Appearance Using Body-Worn IMUs and a Nearby Camera View for Collaborative Egocentric Telepresence

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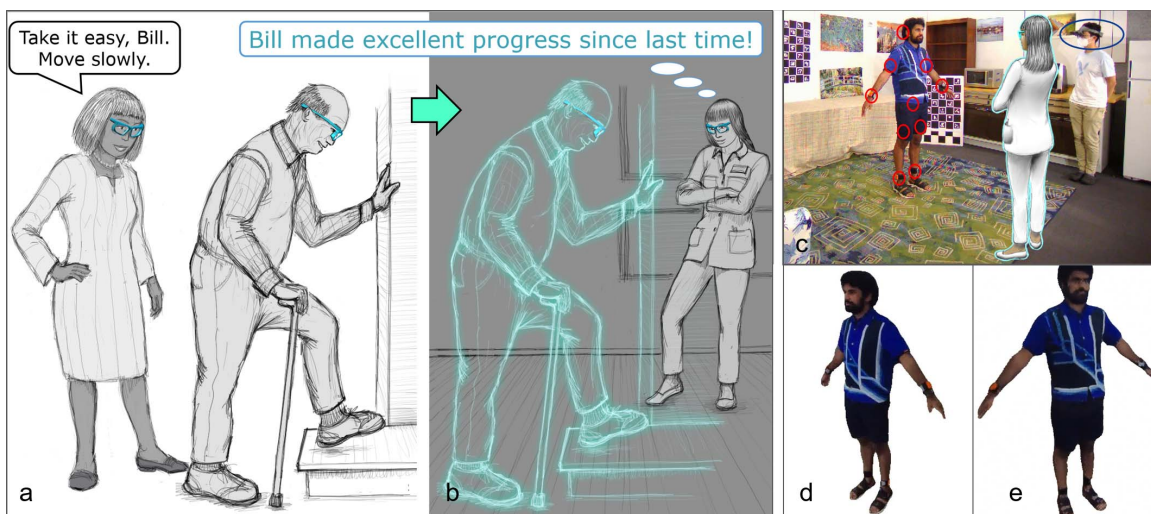


Figure 1: **Future scenario:** a) Patient outside his home, with his pose and appearance captured by his own AR glasses and IMUs, plus camera imagery from the AR glasses of his care partner. b) Patient's data being reviewed by his physical therapist. **Current scenario:** c) Overview of "home" with the "patient" on the left wearing IMUs (circled in red), the "care partner" on the right wearing an AR device with a camera (circled in blue), and a "therapist" who can view the reconstruction remotely using VR glasses. d) Segmented input image from the camera worn by the "care partner." e) Output image for the novel viewpoint of the "therapist."

ABSTRACT

We envision a future in which telepresence is available to users any-time and anywhere, enabled by sensors and displays embedded in accessories worn daily, such as watches, jewelry, belt buckles, shoes, and eyeglasses. We present a collaborative approach to 3D reconstruction that combines a set of IMUs worn by a target person with an external view from another nearby person wearing an AR headset, used for estimating the target person's body pose and reconstructing their appearance, respectively.

Index Terms: Computing methodologies—Artificial intelligence—Computer vision—Reconstruction; Computing methodologies—Machine learning—Machine learning approaches—Neural networks

1 APPROACH

Our goal is to capture the pose and appearance of a target person over time using a set of IMUs worn by the target person

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combined with an external view from another nearby person wearing an AR headset. Toward that end, we have devised a collaborative reconstruction system that consists of 10 *Xsens MTw Awinda* IMUs and an *Ximmerse Rhino X Pro* AR headset. And in the future, we hope to incorporate even fewer IMUs (4) by having the patient also wear an AR headset, with both conventional forward- and downward-looking cameras [2]. Our results demonstrate that integrating the IMUs and an external camera yield better reconstructions than when using either of them alone.

We have developed our approach¹ to support the scenario shown in Figure 1 (right), in which a patient performing physical therapy exercises is captured and a physical therapist can either review their performance or provide interactive feedback using a VR headset. We combine the flexibility of modeling human motion using the *SMPL* [4] body model with the transformation capabilities of image translation and inpainting of neural networks as shown in Figure 2.

Our approach takes as inputs IMU-based body poses and images from the reference view. The wearer's body pose is used to compute the parameters of a *SMPL* model that is fitted in a pre-scan step. We also fit an extra global offset T_{offset} for the *SMPL* template using the body silhouette from reference view. The resulting 3D mesh is used together with images from the reference view at each frame to obtain

¹Code available at: <https://github.com/qzane/CoEgoRecon>

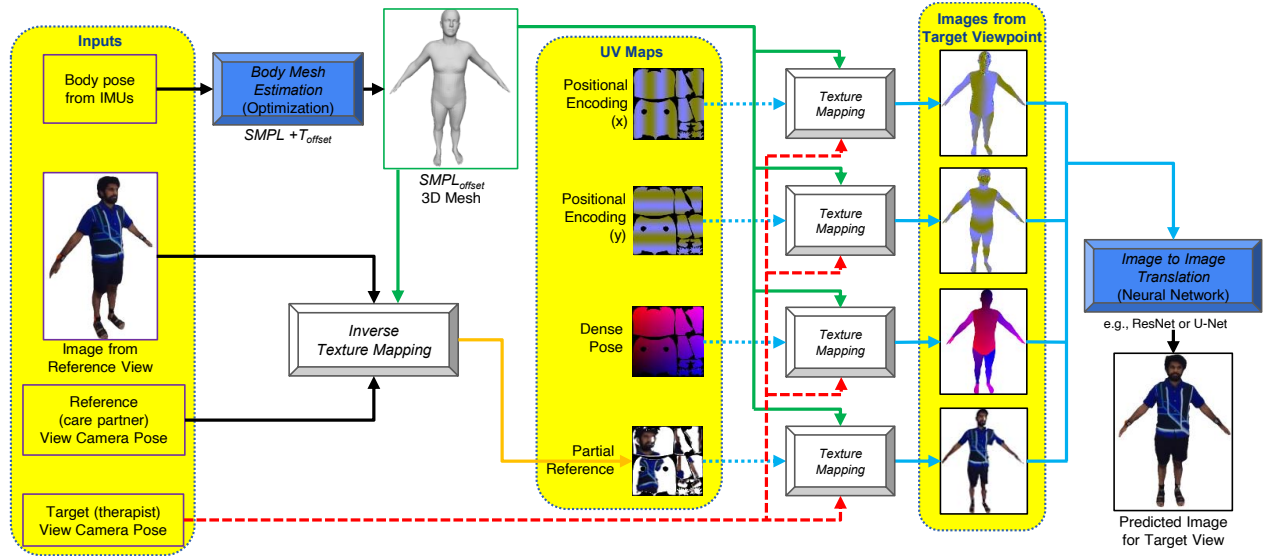


Figure 2: **Overview of our approach.** The body pose from IMUs is used to generate the *SMPL* 3D mesh. The image from the reference view is used together with the 3D mesh to generate a partial reference *uv* map through inverse texture mapping. We then use texture mapping to render the positional encoding (PE), DensePose and partial reference *uv* maps from the target viewpoint. Lastly, we use the rendered PE, DensePose and partial reference images as the input for an image-to-image translation network. The output is the predicted image for the target view.

partial reference *uv* maps via inverse texture mapping. Next, the 3D mesh is used with the partial reference *uv* maps to render partial reference images from the target viewpoint using texture mapping. Similarly, *DensePose* and positional encoding (PE) *uv* maps are rendered in the target view using texture mapping. The rendered images are fed into a neural net (ResNet) that outputs the final image in the target view. The neural net is trained using randomly sampled frames from videos captured by the prototype headset.

2 EXPERIMENTS

We compared our method with a *SMPL* model-based method: Octopus [1], and two model-free methods: Anim-NeRF [3] and HumanNeRF [5]. We use peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM) and learned perceptual image patch similarity (LPIPS) for evaluation. A higher PSNR and SSIM or lower LPIPS usually means the output image is more similar to the ground truth image.

The quantitative evaluation results are shown in Table 1 and the visual comparison is shown in Figure 3. We note that the model-based method generates a blurry texture map, while the model-free methods generate noticeable “ghosting” artifacts with complex motion. Our method has been successful in generating images with clear textural, while effectively minimizing the presence of artifacts.

Evaluation	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Ours	23.748	0.948	0.057
Octopus [1]	14.902	0.877	0.162
Anim-NeRF [3]	18.744	0.897	0.153
HumanNeRF [5]	14.481	0.809	0.221

Table 1: Quantitative evaluation results

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Figure 3: Visual Results: Octopus, Anim-NeRF, HumanNeRF, Ours

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